

Week 1) Book exercise 1.3)

The description in the book translates to a (column-wise) probability matrix with cells

	# fruits		
	# total fruits per box/colour		

as

fruits\box	r (red)	b (blue)	g (green)
a (apples)	$\frac{3}{10}$	$\frac{1}{2}$	$\frac{3}{10}$
o (oranges)	$\frac{4}{10}$	$\frac{1}{2}$	$\frac{3}{10}$
c (cimes)	$\frac{3}{10}$	$\frac{0}{2}$	$\frac{4}{10}$

It is trivial, but you should check that each column sums to 1 (i.e. 100%).

We also have a probability distribution over the boxes given by $p(r)=0.2$, $p(b)=0.2$, $p(g)=0.6$

(this also sums to 1).

Using the law of total probability, the probability of selecting an apple is given by

$$\begin{aligned} p(a) &= p(a|r) \cdot p(r) + p(a|b) \cdot p(b) + p(a|g) \cdot p(g) \\ &= \frac{3}{10} \cdot 0.2 + \frac{1}{2} \cdot 0.2 + \frac{3}{10} \cdot 0.6 \\ &= 0.34. \end{aligned}$$

Based on Bayes' theorem, the probability of the green box given an orange is

$$\frac{1}{10}$$

$$p(g/o) = \frac{p(o/g) \cdot p(g)}{p(o)}$$

where $p(o)$ (by the law of total probability) is given by

$$\begin{aligned} p(o) &= p(o/r) \cdot p(r) + p(o/b) \cdot p(b) + p(o/g) \cdot p(g) \\ &= \frac{4}{10} \cdot 0.2 + \frac{1}{2} \cdot 0.2 + \frac{3}{10} \cdot 0.6 \\ &= 0.36. \end{aligned}$$

Thus, we have

$$\begin{aligned} p(g/o) &= \frac{p(o/g) \cdot p(g)}{p(o)} \\ &= \frac{\frac{3}{10} \cdot 0.6}{0.36} \\ &= 0.5. \end{aligned}$$

Book exercise 1.7)

We want to show $\int_{-\infty}^{\infty} N(x|\mu, \sigma^2) dx = 1$

where $N(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2\sigma^2}(x-\mu)^2\right\}$

denotes the normal density.

We first let $I = \int_{\mathbb{R}} \exp\left\{-\frac{1}{2\sigma^2} x^2\right\} dx$

and note

$$\begin{aligned} I^2 &= II = \left(\int_{\mathbb{R}} \exp\left\{-\frac{1}{2\sigma^2} x^2\right\} dx \right) \left(\int_{\mathbb{R}} \exp\left\{-\frac{1}{2\sigma^2} y^2\right\} dy \right) \\ &= \int_{\mathbb{R}} \int_{\mathbb{R}} \left(\exp\left\{-\frac{1}{2\sigma^2} x^2\right\} \exp\left\{-\frac{1}{2\sigma^2} y^2\right\} \right) dx dy \\ &= \int_{\mathbb{R}} \int_{\mathbb{R}} \underbrace{\exp\left\{-\frac{1}{2\sigma^2} x^2 - \frac{1}{2\sigma^2} y^2\right\}}_{(*)} dx dy \end{aligned}$$

Recall that we transform to polar coordinates (r, θ) by setting $x = r \cdot \cos(\theta)$, $y = r \cdot \sin(\theta)$, $r^2 = x^2 + y^2$.

Furthermore, if f is a continuous function on a polar rectangle $R = \{(r, \theta) \mid a \leq r \leq b, \alpha \leq \theta \leq \beta\}$ where $0 \leq r$, $0 \leq \beta - \alpha \leq 2\pi$, then

$$\iint f(x, y) dA = \int_{\alpha}^{\beta} \int_a^b f(r \cdot \cos(\theta), r \cdot \sin(\theta)) \cdot r dr d\theta.$$

Often forgotten!

Now, consider (*)

$$\begin{aligned}\exp\left\{-\frac{1}{2\sigma^2}x^2 - \frac{1}{2\sigma^2}y^2\right\} &= \exp\left\{-\frac{1}{2\sigma^2}(x^2 + y^2)\right\} \\ &= \exp\left\{-\frac{1}{2\sigma^2}r^2\right\}.\end{aligned}$$

We are integrating over the entire plane so

$$R = \{(r, \theta) \mid 0 \leq r \leq \infty, 0 \leq \theta \leq 2\pi\},$$

so, we have $a=0, b=\infty, \alpha=0, \beta=2\pi$.

We now see that

$$\begin{aligned}I^2 &= \int_{\mathbb{R}} \int_{\mathbb{R}} \exp\left\{-\frac{1}{2\sigma^2}x^2 - \frac{1}{2\sigma^2}y^2\right\} dx dy \\ &= \int_0^{2\pi} \int_0^{\infty} \exp\left\{-\frac{1}{2\sigma^2}r^2\right\} \cdot r dr d\theta\end{aligned}$$

Recall integration by substitution,

$$\int_a^b f(g(x)) g'(x) dx = \int_{u(a)}^{u(b)} f(u) du$$

where $u=g(x)$, $du=g'(x)dx$.

We set $u = -\frac{r^2}{2\sigma^2}$ with $\frac{du}{dr} = -\frac{r}{\sigma^2}$

$$= -\sigma^2 \int_0^{2\pi} \int_0^{\infty} \exp\left\{-\frac{1}{2\sigma^2}r^2\right\} \frac{-r}{\sigma^2} dr d\theta$$

$$= -\sigma^2 \int_0^{2\pi} \int_0^{-\infty} \exp\{u\} du d\theta$$

$$= -\sigma^2 \int_0^{2\pi} [\exp(i\theta)]_0^{-\infty} d\theta$$

$$= -\sigma^2 \int_0^{2\pi} (0 - 1) d\theta$$

$$= \sigma^2 \int_0^{2\pi} 1 d\theta$$

$$= 2\sigma^2\pi$$

This was a lot of math to show

$$I^2 = 2\sigma^2\pi \Rightarrow I = \sqrt{2\sigma^2\pi}.$$

But we actually want to show that

$$\int_{\mathbb{R}} \mathcal{N}(x|\mu, \sigma) dx = \int_{\mathbb{R}} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2\sigma^2} (x-\mu)^2\right\} dx$$

$$= \frac{1}{\sqrt{2\pi\sigma^2}} \underbrace{\int_{\mathbb{R}} \exp\left\{-\frac{1}{2\sigma^2} (x-\mu)^2\right\} dx}_{I = \sqrt{2\sigma^2\pi}}$$

$$= \frac{1}{\sqrt{2\pi\sigma^2}} \cdot \sqrt{2\sigma^2\pi}$$

$$= \underline{1}$$

Book exercise 1.10 (extended)

Let $a, b \in \mathbb{R}$ be constants and let X, Y be random variables.

Note that we implicitly assume $E[X]$, $E[Y]$, $\text{Var}(X)$ and $\text{Var}(Y)$ to be finite.

We will assume X and Y to be continuous

- you are welcome to go through the exercise when X and/or Y are discrete (the approach is exactly the same),
- if you are a lazy mathematician/measure theorist, you simply note that summation is integration with a different measure.

We now have:

(1) By linearity of the integral

$$\begin{aligned} E[a + bX] &\stackrel{\text{def.}}{=} \int_{\mathbb{R}} (a + bx) p(x) dx \\ &= a \underbrace{\int_{\mathbb{R}} p(x) dx}_{=1} + b \cdot \int_{\mathbb{R}} x p(x) dx \\ &= a + b \cdot E[X], \end{aligned}$$

(2) and further

$$\begin{aligned} E[X + Y] &= \iint_{\mathbb{R} \times \mathbb{R}} (x + y) p(x, y) dx dy \\ &= \iint_{\mathbb{R} \times \mathbb{R}} x p(x, y) dx dy + \iint_{\mathbb{R} \times \mathbb{R}} y p(x, y) dx dy \end{aligned}$$

$$= \iint_{\mathbb{R} \times \mathbb{R}} x p(x, y) dy dx + \iint_{\mathbb{R} \times \mathbb{R}} y p(x, y) dx dy$$

$$= \int_{\mathbb{R}} x \left(\int_{\mathbb{R}} p(x, y) dy \right) dx + \int_{\mathbb{R}} y \left(\int_{\mathbb{R}} p(x, y) dx \right) dy$$

$p(x)$ $p(y)$

$$= \int_{\mathbb{R}} x p(x) dx + \int_{\mathbb{R}} y p(y) dy$$

$$\stackrel{\text{def.}}{=} E[X] + E[Y],$$

3) We can now show (using

$$\text{Var}(X) \stackrel{\text{def.}}{=} E[(X - E[X])^2]$$

← This is just a constant.

$$= E[X^2 + E[X]^2 - 2X E[X]]$$

$$\stackrel{(1)+(2)}{=} E[X^2] + E[X]^2 - 2 E[X] E[X]$$

$$= E[X^2] - E[X]^2,$$

4) and

$$\text{Var}(a + bX) \stackrel{\text{def.}}{=} E[(a + bX - E[a + bX])^2]$$

$$\stackrel{(1)+(2)}{=} E[(a + bX - a - bE[X])^2]$$

$$= E[b^2 (X - E[X])^2]$$

$$\stackrel{\text{def.}+(2)}{=} b^2 \text{Var}(X),$$

(5) Furthermore, we have

$$\text{Var}(X+Y) \stackrel{\text{def.}}{=} E[(X+Y) - E[X+Y]]^2]$$

$$\stackrel{(1)}{=} E[(X - E[X]) + (Y - E[Y])]^2]$$

$$= E[(X - E[X])^2 + (Y - E[Y])^2 + 2(X - E[X])(Y - E[Y])]$$

$$\stackrel{(1)+(2)}{=} E[(X - E[X])^2] + E[(Y - E[Y])^2]$$

$$+ 2E[(X - E[X])(Y - E[Y])]$$

$$\stackrel{\text{def.}}{=} \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y),$$

(6) and

$$\text{Cov}(X, Y) \stackrel{\text{def.}}{=} E[(X - E[X])(Y - E[Y])]$$

$$= E[XY - XE[Y] - YE[X] + E[X]E[Y]]$$

$$\stackrel{(1)+(2)}{=} E[XY] - E[X]E[Y] - E[Y]E[X]$$

$$+ E[X]E[Y]$$

$$= E[XY] - E[X]E[Y],$$

We now assume that X and Y are independent (often written as $X \perp\!\!\!\perp Y$ in sloppy notation - please don't do this at the exam!).

(7) Then

$$\text{Cov}(X, Y) \stackrel{(6)}{=} E[X \cdot Y] - E[X]E[Y]$$

$$\stackrel{\text{def.}}{=} \iint_{\mathbb{R} \times \mathbb{R}} (x \cdot y) p(x, y) dx dy - E[X]E[Y]$$

$$\stackrel{\text{by (1)}}{=} \iint_{\mathbb{R} \times \mathbb{R}} x \cdot y \cdot p(x) \cdot p(y) dx dy - E[X]E[Y]$$

$$= \left(\int_{\mathbb{R}} x p(x) dx \right) \left(\int_{\mathbb{R}} y p(y) dy \right) - E[X]E[Y]$$

$$\stackrel{\text{def.}}{=} E[X]E[Y] - E[X]E[Y]$$

$$= 0$$

(8) and

$$\begin{aligned} \text{Var}(X+Y) &\stackrel{(5)}{=} \text{Var}(X) + \text{Var}(Y) + 2 \underbrace{\text{Cov}(X, Y)}_{=0} \\ &= \text{Var}(X) + \text{Var}(Y) \end{aligned}$$

To see that (7) is not a two-way statement (i.e., $\text{Cov}(X, Y) = 0 \not\Rightarrow X \perp\!\!\!\perp Y$) consider two random variable X, Y with a joint probability distribution given by the table

$X \backslash Y$	-1	0	1
-1	0	$\frac{1}{4}$	0
0	$\frac{1}{4}$	0	$\frac{3}{4}$
1	0	$\frac{3}{4}$	0

and marginal distributions

$$p(X=-1) = \frac{1}{4}, \quad p(X=0) = \frac{1}{2}, \quad p(X=1) = \frac{1}{4},$$

$$p(Y=-1) = \frac{1}{4}, \quad p(Y=0) = \frac{1}{2}, \quad p(Y=1) = \frac{1}{4}.$$

It is trivial that

$$E[X] \stackrel{\text{def.}}{=} -1 \cdot \frac{1}{4} + 0 \cdot \frac{1}{2} + 1 \cdot \frac{1}{4} = 0,$$

$$E[Y] \stackrel{\text{def.}}{=} -1 \cdot \frac{1}{4} + 0 \cdot \frac{1}{2} + 1 \cdot \frac{1}{4} = 0,$$

so

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

$$= E[XY]$$

$$\stackrel{\text{def.}}{=} \begin{aligned} &((-1) \cdot (-1) \cdot 0) + ((-1) \cdot 0 \cdot \frac{1}{4}) + ((-1) \cdot 1 \cdot 0) \\ &+ (0 \cdot (-1) \cdot \frac{1}{4}) + (0 \cdot 0 \cdot 0) + (0 \cdot 1 \cdot \frac{1}{4}) \\ &+ (1 \cdot (-1) \cdot 0) + (1 \cdot 0 \cdot \frac{1}{4}) + (1 \cdot 1 \cdot 0) \end{aligned}$$

$$= 0$$

But $X \not\perp Y$. Indeed, we see that

$$\begin{aligned} 0 &\stackrel{\text{By construction}}{=} p(X=-1, Y=-1) \neq p(X=-1) \cdot p(Y=-1) \\ &= \frac{1}{4} \cdot \frac{1}{4} \\ &= \frac{1}{16}. \end{aligned}$$